

EMG-driven hand model based on the classification of individual finger movements

Maria V. Arteaga^a, Jenny C. Castiblanco^a, Ivan F. Mondragon^b, Julian D. Colorado^c, Catalina Alvarado-Rojas^{c,*}

^a School of Engineering, Pontificia Universidad Javeriana, Cr 7 No. 40-62, Bogota, Colombia

^b Department of Industrial Engineering, Pontificia Universidad Javeriana, Cr 7 No. 40-62, Bogota, Colombia

^c Department of Electronics Engineering, Pontificia Universidad Javeriana, Cr 7 No. 40-62, Bogota, Colombia

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ABSTRACT

The recovery of hand motion is one of the most challenging aspects in stroke rehabilitation. This paper presents an initial approach to robot-assisted hand-motion therapies. Our goal was twofold: firstly, we have applied machine learning methods to identify and characterize finger motion patterns from healthy individuals. To this purpose, Electromyographic (EMG) signals have been acquired from flexor and extensor muscles in the forearm using surface electrodes. Time and frequency features were used as inputs to machine learning algorithms for recognition of six hand gestures. In particular, we compared the performance of Artificial Neural Networks (ANN), Support Vector Machines (SVM) and k-Nearest Neighbor (k-NN) algorithms for classification. Secondly, each identified gesture was turned into a joint reference trajectory by applying interpolation methods. This allowed us to reconstruct the hand/finger motion kinematics and to simulate the dynamics of each motion pattern. Experiments were carried out to create an EMG database from 20 control subjects, and a VICON camera tracking system was used to validate the accuracy of the proposed system. The average correlation between the EMG-based generated joint trajectories and the tracked hand-motion was 0.91. Furthermore, statistical analysis applied to 14 different SVM, ANN and k-NN configurations showed that Fine k-NN and Weighted k-NN have a better performance for the classification of gestures (98% of accuracy). In a future, the trajectories controlled by EMG signals could be applied to an exoskeleton or hand-robotic prosthesis for rehabilitation.

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1. Introduction

Humans use their hands for a large number of Activities of Daily Living (ADL), being a fundamental tool to manipulate objects and interact with the environment. Hands are composed by 29 skeletal muscles, 27 bones, and 15 joints, allowing up to 21 degrees of freedom [1–4]. The sophisticated hand movements are the result of a complex processing in the central nervous system and allow to execute diverse ADL by means of basic movements such as power or precision grasps (i.e. feeding, driving, writing) [5,6]. For this reason, alterations of the hand function greatly affect how the individual gets along ADL. In particular, hand function can be seriously affected after stroke, which affects around 100–200/100,000

of world population per year, and is the main cause of motor disability [7,8]. Furthermore, nearly 80% of patients present upper limb impairment after a stroke event [9]. The impediment to perform different activities greatly affects the independence of the patient and can produce long-term disability.

Rehabilitation therapy is a common treatment after stroke and is considered essential for recovery. For instance, around 70% of the motor function is recovered between 3 and 6 months after a stroke event [10]. However, the recovery process can be slower in the upper extremity compared to the lower extremity or cognitive function [11], partly due to the complexity of the upper extremity movements [12]. Hand therapies include repetitive task-specific training, in order to repair finger muscle weakness and spasticity, somatosensory loss, and abnormal coactivation of finger muscles during isolated movements [13,14].

Recent advances in hand rehabilitation have suggested that robotic-assisted therapy could help in the recovery of motor function [15–18,4,19]. Specifically, it could increase the intensity and frequency of the therapy, provide an accurate control of the motor assistance, and make an objective measurement of the patient

* Corresponding author.

E-mail addresses: arteagam@javeriana.edu.co (M.V. Arteaga), jenny.castiblanco@javeriana.edu.co (J.C. Castiblanco), imondragon@javeriana.edu.co (I.F. Mondragon), coloradoj@javeriana.edu.co (J.D. Colorado), catalina.alvarado@javeriana.edu.co (C. Alvarado-Rojas).

progress [20,4]. Mainly two types of robotic systems have been developed for this purpose, the end-effector and the exoskeleton [21]. The end-effector system provides external assistance to the movement by applying forces at the end of the digits [22]. Some popular systems used in hand rehabilitation are the Amadeo (Tyromotion GmbH, Austria) [23,24] and the ReHapticKnob [25,26]. Conversely, the exoskeleton devices support the hand and fingers, providing a direct control to the hand joints. The function of exoskeleton ranges from assistance of broad grasp and knob movements such as the HWARD and Haptic-Knob devices [27,28], to highly specialized devices that control up to 20 DOF of finger and hand movements, such as the CyberGrasp (CyberGlove Systems LLC), the Hand-of-Hope (Rehab-Robotics Comp.) [29], the HEXXOR [22], the Soft RoboticGlove [30], the GloReha [31] and the ReHand [32]. For a detailed review of technologies for hand assisted rehabilitation see [33,34,4]. Despite the growing interest to develop robotic systems in the last years, a significant improvement of hand function after robotic rehabilitation still presents few evidences and requires further studies, specially because of the low number of patients and the variability of the robotic-systems [21,4,19].

The control system of an exoskeleton can be done by mechanical or electrical signals. Mechanical signals include finger or joint positions, forces and torques, voice commands, or switches. Electrical signals from the body itself, can also drive the system response. In particular, Electroencephalography (EEG) and Electromyography (EMG) signals reflect the patient's intention and control of movement [35–37]. The EMG signal represents the electrical activity of the muscle sent from the central nervous system when a contraction is desired. The myoelectrical signals can be acquired by invasive and non-invasive electrodes, the latter being widely used in rehabilitation. For instance, the intention to flex and extend the fingers can be partially recorded using surface electrodes (sEMG) in both forearm muscles, the Flexor Digitorum Superficialis and the Extensor Digitorum Communis [30], respectively. However, due to the low amplitude of the EMG signal 1–10 mV and the interference from other muscle activity (cross-talk), the EMG signal is highly affected by noise, requiring a high quality analog acquisition and advanced digital processing algorithms. Furthermore, in order to control the exoskeleton, a simultaneous and proportional control of several DOFs is also necessary [38]. These challenges have limited the current application of sEMG signals to exoskeleton systems in clinical and commercial applications [38,21,36]. Indeed, from more than 165 existing devices reported in [33], 30% have 5 or more DOFs, and only 10% have more than 10 DOFs. From those only 10% use EMG signals as the control signal [39,40,32].

Thus, robotic-rehabilitation approaches require developing robust machine learning methods to detect the hand movements and control the exoskeleton in real time. In the last 10 years, different studies have explored the identification of hand movements from EMG signals on the forearm, with accuracies above 90%. In one of the first representative studies, Tenore et al. (2007) differentiated 12 movements of the hand and wrist, from EMG signals recorded with a 32-channel system. Posterior studies have detected hand movements using features in time, frequency and time-frequency domains. In the time domain, the most used features are: Mean Absolute Value (MAV), Willison Amplitude (WAMP), Waveform Length (WL), Zero-Crossing (ZC) [41–43], but others such as entropy [44] and Auto-Regressive coefficients (AR) [45–47] have been used with satisfactory results [48,49]. From the frequency domain, measures from the Power Spectral Density (PSD) have been extracted, including mean frequency and frequency ratios [50,51,45]. Finally, features in the time-frequency domain have also been used, such as Short-Time Fourier Transform (STFT) and Wavelet coefficients [52,53,45]. Even though time features have been suggested to perform better than frequency, or time-frequency features, they contain a high redundancy [48,49]. For this reason, dimensionality

reduction techniques, such as PCA, have been applied to improve the results (see [49] for a review). On the other hand, automatic classification methods have been also compared in previous studies, with a predominance of Linear Discriminant Analysis (LDA) that has accuracies above 96% for more than 10 channels [45,54], Artificial Neural Networks (ANN) with accuracies between 88% for 4 movements, 3 subjects and 2 channels, to 99% for 5 movements, 1 subject and 5 channels [55–58], Hidden Markov Models with accuracy of 80% for 12 movements and 16 channels [59], Support Vector Machines (SVM) with accuracy of 98% for 15 movements and more than 10 channels [46,60], and finally k-Nearest Neighbors (kNN) with accuracy of 89% for 7 movements, 8 channels, and 5 subjects [66]. Whereas the results of classification of finger movements seems promising, many of the previous studies used multi-channel systems with more than 10 electrodes, or decode broad hand instead of individual finger movements [46,49].

Therefore, in this study we proposed a machine learning approach to decode 6 hand gestures (5 individual finger flexion/extension and 1 hand closed) with 4 electrodes in the forearm. For this purpose, we have analyzed EMG signals from control volunteers, and performed feature extraction, as well as application and comparison of SVM, ANN, and k-NN classifiers. Additionally, we have used the classification of finger movements to define the joint trajectories of a virtual hand model. This model was developed as a rigid body with ramified joints, with the kinematics and dynamics following Newton-Euler formulation. The overall architecture of the system can be seen in Fig. 1. The main motivation of this study was to develop a robust classification of 6 hand movements, to drive the movement of a virtual hand model, with a low number of EMG channels and a high number of DOFs. The virtual hand model could be replaced by an exoskeleton in the future.

2. Methods

This section presents the proposed methods for the three main phases of this study: EMG signal acquisition and pre-processing, Machine Learning for gesture recognition, and Robotic-hand dynamics modeling for motion pattern analysis.

2.1. EMG signal acquisition and pre-processing

Fig. 2 details the experimental protocol for EMG signal acquisition during 6 different hand gestures to be automatically classified by the system. The EMG signals were recorded from 20 healthy volunteers (all females, mean age 20 ± 1.9 years old), who reported no previous stroke events or hand impairments. They agreed to participate in the study and accepted the informed consent. Based on previous works, four superficial electrodes (Ag/AgCl) were placed in four muscles around the forearms: extensor digitorum, extensor carpi ulnaris, flexor digitorum superficialis and flexor carpi ulnaris [61,49]. Additionally, one electrode was placed in the elbow for reference, as it is considered an electrically neutral tissue [61]. Ideally, the electrodes should be placed at the muscle belly around 1–2 cm from each other [62]. However, in order to obtain EMG signals that properly reflect the activity of different movements, the two electrodes placed along the flexor muscles needed to be slightly shifted for each gesture. The gestures of interest (6), corresponded to individual finger flexion/extension (5) and hand closed (1). During the experiment the volunteer was asked to perform 20 repetitions for each of the 6 gestures, resting for 3 s between repetitions, and 30 s between gestures.

For the acquisition and digitization of the signals, the commercial MyoWare Muscle Sensor, National Instruments DAQ (NI-USB 6366) and SignalExpress software were used. A previous characterization of the sensor was required since the data-sheet did

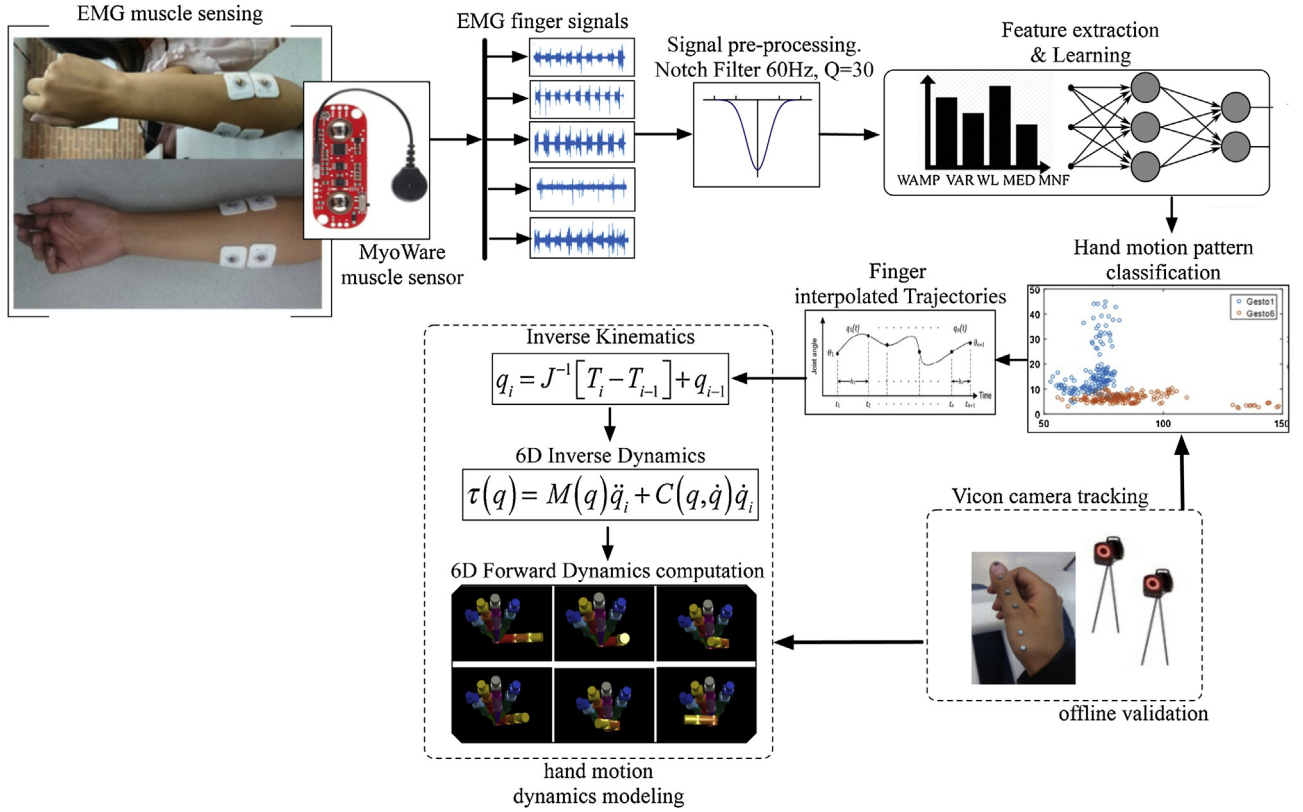


Fig. 1. System architecture. EMG signals were used as inputs to the machine learning module to accurately extract finger motion patterns that could be further applied to an exoskeleton or hand-robotic prosthesis for rehabilitation scenarios. We used spatial dynamics algorithms to validate how these finger motion patterns could be used as input joint trajectories in robotic-assisted therapies.

not provide detailed information of the sensor specifications. The obtained characteristics of the sEMG sensors were: maximal gain of 400 V/V, bandwidth of 416 Hz, input impedance of 110 GΩ and CMRR of 110 dB. The EMG signals were digitized with a sampling frequency of 2 kHz and a 16-bits ADC. The digital signals were processed in Matlab (The Mathworks, Natick, MA, USA), by applying two digital filters: a notch filter at 60 Hz (Butterworth, 10th order) and a bandpass filter between 5 Hz and 300 Hz (Butterworth, 32nd order). In order to fix the phase distortion created by the filter, a zero-phase filtering strategy was applied. The frequency response of the implemented filters corresponds to the characteristics of the EMG signal, where most of the information lies within the 50–150 Hz band [49].

2.2. Machine learning for gesture recognition

A machine learning approach was used to identify patterns in the EMG signals of 6 different hand gestures. First of all, a feature extraction stage was implemented, in order to obtain relevant characteristics from the signals. Once the EMG signal was filtered, we computed six features in time and frequency domains: MAV, WAMP (threshold of 5 mV) [67], VAR, WL, MDF and MNF, as detailed in Table 1.

The signal acquired when the subject was executing a repetition of a gesture was called an *event*. Each subject had 20 events per gesture, each event lasts 3 s. Overall, we had 2400 events for the entire experiment, with 400 events per gesture (20 subjects × 20 repetitions). From each event, we extracted the 6 features mentioned above for the 4 electrodes that were simultaneously acquired, giving a total of 24 features per event. Table 2, shows the convention used for feature identification.

Table 1

Time and frequency features extracted from the EMG signals.

Domain	Feature	Formulation
Time	Mean Absolute Value (MAV)	$\frac{1}{N} \sum_{i=1}^N x_i $
Time	Willison amplitude (WAMP)	$\sum_{i=1}^N f(x_i - x_{i+1})$ $f(x) = \begin{cases} 1 & x \geq 5 \text{ mV} \\ 0 & x < 5 \text{ mV} \end{cases}$
Time	Variance (VAR)	$\frac{1}{N-1} \sum_{i=1}^N x_i^2$
Time	Waveform length (WL)	$\sum_{i=1}^N x_{i+1} - x_i $
Frequency	Median Frequency (MDF)	$\sum_{j=1}^{MDF} PSD_j = \sum_{j=MDF}^M PSD_j$
Frequency	Mean frequency (MNF)	$\frac{\sum_{j=1}^M f_j PSD_j}{\sum_{j=1}^M PSD_j}$

Table 2

Notation used for the 24 features extracted: 6 features × 4 channels.

	Channel 1	Channel 2	Channel 3	Channel 4
MAV	Feat1	Feat2	Feat3	Feat4
WAMP	Feat5	Feat6	Feat7	Feat8
VAR	Feat9	Feat10	Feat11	Feat12
WL	Feat13	Feat14	Feat15	Feat16
MDF	Feat17	Feat18	Feat19	Feat20
MNF	Feat21	Feat22	Feat23	Feat24

From the feature space, we trained three types of classification algorithms for the 6 gestures of interest: (i) Support Vector Machine (SVM) and (ii) Artificial Neural Network (ANN) and (iii) k-nearest neighbors (k-NN). The optimization of these supervised algorithms, required an initial training stage with 70% of the events, and a testing stage with the remaining 30%. Moreover, we implemented different configurations for the classifiers, in order to find

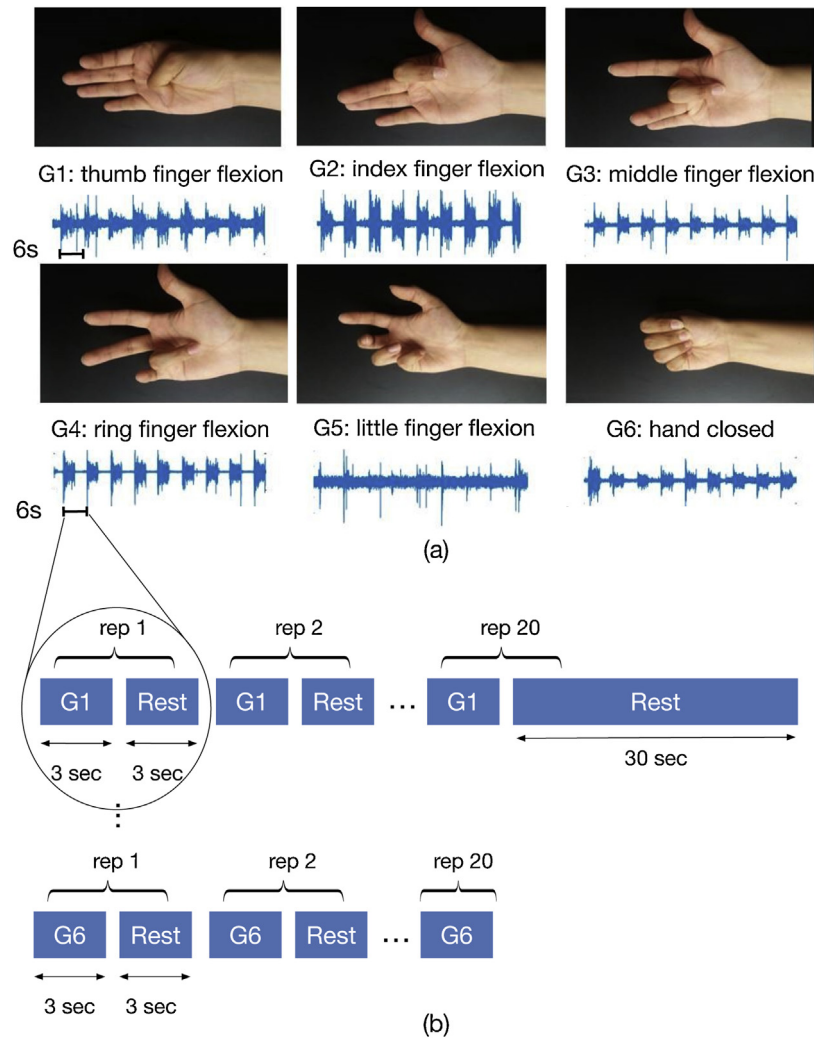


Fig. 2. (a) Acquisition setup: each individual performed 6 gestures (G1–G6) based on the flexion of each finger independently, and the closing of the hand. The signals depicted below each gesture correspond to the EMG measured signals. (b) Time diagram of the experimental protocol.

Table 3

Configuration of SVM, ANN and k-NN classifiers.

Classifier	Kernel	Neurons layer 1	Neurons layer 2	Training Function	Activation function	Neighbors	Distance
SVM1	Linear	–	–	–	–	–	–
SVM2	Quadratic	–	–	–	–	–	–
SVM3	Cubic	–	–	–	–	–	–
SVM4	Fine Gaussian	–	–	–	–	–	–
SVM5	Medium Gaussian	–	–	–	–	–	–
ANN1	–	15	–	Levenberg-MarquardtBackpropagation	tan-sigmoid	–	–
ANN2	–	15	8	Levenberg-MarquardtBackpropagation	log-sigmoid	–	–
ANN3	–	15	8	Levenberg-MarquardtBackpropagation	tan-sigmoid	–	–
ANN4	–	15	8	Levenberg-MarquardtBackpropagation	tan-sigmoid(log-sigmoid output function)	–	–
KNN1	–	–	–	–	–	Fine (1)	Euclidean
KNN2	–	–	–	–	–	Medium (10)	Euclidean
KNN3	–	–	–	–	–	Medium (10)	Cosine
KNN4	–	–	–	–	–	Medium (10)	Weighted
KNN4	–	–	–	–	–	Medium (10)	Cubic

the one with the best performance. For the SVM classifier, 5 kernel functions were tested. Additionally, the one-to-one and one-vs-all multi-class strategies were implemented, because SVM is a binary classifier. On the other hand, for the ANN classifier, we varied between 1 or 2 hidden layers with different training algorithms and activation functions. Finally, for the k-NN classifier, we changed the number of neighbors and the distance metric, as detailed in Table 3.

2.3. Dynamics modeling for motion pattern analysis

The hand was modeled as a branched physical system, in which each finger is composed by a chain of rigid bodies serially connected through rotational joint links. Therefore, conventional Denavit-Hartenberg (DH) geometrical parameters were used to represent the morphology of the hand robotic model. In this sense, Equations

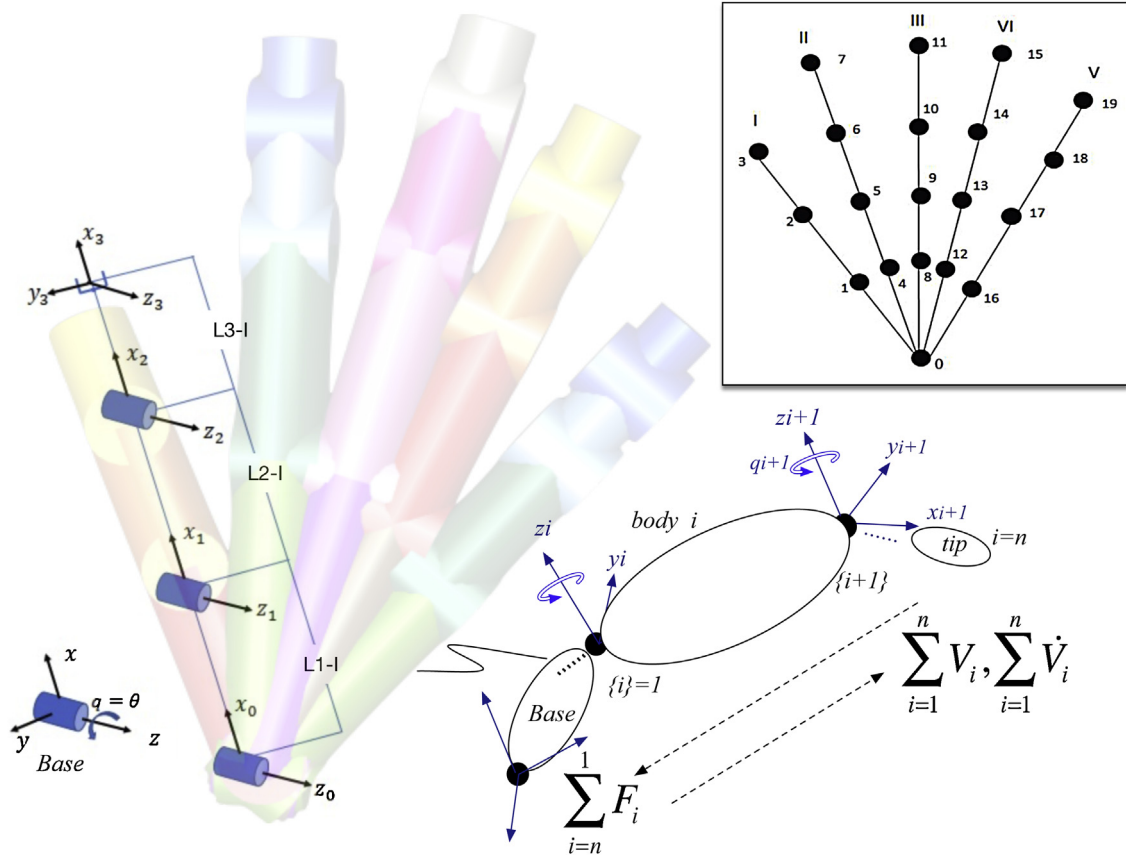


Fig. 3. Rigid multibody diagram to represent the hand robotic system. Spatial velocities V_i and accelerations $\dot{V}_i$ are propagated from the base (carpal bone) to the distal phalange, whereas spatial forces F_i are calculated in a backward recurrence from the distal phalange to the carpal bone. Torques for each joint are obtained from these forces. The inset shows the connection diagram for each serial chain (fingers).

of Motion (EoM) were formulated based on Newton-Euler formalism using spatial operators. Fig. 3 details the multi-body diagram to represent the hand robotic system.

Spatial operators led to six-dimensional physical quantities that combined the angular and linear variables of rigid-body motions. Using spatial algebra notation, velocities V_i , accelerations $\dot{V}_i$, and forces F_i of a rigid body i were expressed with respect to its joint frame $\{i\}$ using six-dimensional vectors, as follows:

$$V_i = \begin{bmatrix} \omega_i \\ v_i \end{bmatrix}, \quad \dot{V}_i = \begin{bmatrix} \dot{\omega}_i \\ \dot{v}_i \end{bmatrix}, \quad F_i = \begin{bmatrix} \tau_i \\ f_i \end{bmatrix} \in \mathbb{R}^{6 \times 1} \quad (1)$$

Where ω_i , $\dot{\omega}_i$ and τ_i represent the corresponding angular motions, and v_i , $\dot{v}_i$ and f_i are the linear variables. We required the joint trajectory angular position, velocity and acceleration for each finger (denoted by q_i , $\dot{q}_i$, $\ddot{q}_i$, respectively) as inputs to the dynamics model. The subscript i denotes the number of the joint in a digit j ($\forall j = I \dots V$). Joint trajectories were generated based on the classified gestures defined in the previous section. Both forward and inverse kinematics methods were applied to relate the motion of each joint based on the defined trajectories q_i . To propagate the physical quantities along the system, the standard homogeneous transformation matrix $A_{i+1,i}$ was used.

$$A_{i+1,i} = \begin{bmatrix} r_{i+1,i} & p_{i,i+1} \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} \cos q_i & -\cos \alpha_i \sin q_i & \sin \alpha_i \sin q_i & a_i \cos q_i \\ \sin q_i & \cos \alpha_i \cos q_i & -\sin \alpha_i \cos q_i & a_i \sin q_i \\ 0 & \sin \alpha_i & \cos \alpha_i & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (2)$$

Table 4
DH parameters.

	Joint	α	a	q	d
Thumb (cf. Fig. 3)	1	0	L_{1-I}	$q_1(t)$	0
	2	0	L_{2-I}	$q_2(t)$	0
	3	0	L_{3-I}	$q_3(t)$	0
Digits II-V ($j = II \dots V$)	1	$-\pi/2$	L_{1-dig_j}	$q_{1,dig_j}(t)$	0
	2	0	L_{2-dig_j}	$q_{2,dig_j}(t)$	0
	3	0	L_{3-dig_j}	$q_{3,dig_j}(t)$	0
	4	0	L_{4-dig_j}	$q_{4,dig_j}(t)$	0

The term $r_{i+1,i} \in \mathbb{R}^{3 \times 3}$ is the rotation matrix that relates the joint frame $\{i+1\}$ with $\{i\}$ and $p_{i,i+1} \in \mathbb{R}^3$ is the position vector that connects consecutively frames. Both terms are dependent on the DH parameters α_i , a_i , d_i (cf. Table 4). By using $r_{i+1,i}$ and $p_{i,i+1}$ from Eq. (2), six-dimensional operators for rotation R and translation P are defined in Eq. (3).

$$P = \begin{bmatrix} U & \tilde{p}_{i,i+1} \\ 0 & U \end{bmatrix}, \quad \in \mathbb{R}^{6 \times 6} \quad (3)$$

$$R = \begin{bmatrix} r_{i+1,i} & 0 \\ 0 & r_{i+1,i} \end{bmatrix} \in \mathbb{R}^{6 \times 6}$$

By using R and P from Eq. (3), the EoM in Eq. (1) can be applied to each segment of the fingers, and propagated from carpal to metacarpals and along the phalanges: proximal, intermediate and distal. To properly convert the current three-dimensional form of r and p into six-dimensional operators, we use the skew-symmetric

matrix of the position vector p , denoted as $\tilde{p}_{i,i+1}$ in Eq. (3). The term U refers to the identity operator.

The complete expressions for calculating the EoM: spatial velocities V_i , accelerations $\dot{V}_i$, forces F_i and moments τ_i are defined in Algorithm 1. The inputs of the algorithm are the joint trajectories defined by the user ($q_i, \dot{q}_i, \ddot{q}_i$), whereas the output is the required torque τ_i for each joint i .

The method is divided into two recurrences: (i) a forward calculation and propagation of the spatial velocities and accelerations and (ii) a backward recurrence to calculate spatial forces and the torques acting on each joint of the fingers. In Algorithm 1, the equation labeled as (1) refers to the expression of the spatial velocity. As mentioned, $\dot{q}_i$ is the angular speed (rad/s) profile for each joint i along the fingers. The term H allows for the projection of $\dot{q}_i$ onto the axis of motion. Since all coordinate frames were placed by following the DH method, the rotational motion of each joint applied around the z_i -axis, hence, $H = \begin{bmatrix} 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix}^T$. By differentiating (1) in Algorithm 1 with respect to time, the spatial accelerations are obtained (labeled as equation (2)). Finally, the spatial forces (labeled as (3)) are calculated by taking into account the spatial inertia of each segment of the finger, denoted as $I_i \in \mathbb{R}^{6 \times 6}$.

Algorithm 1. Inverse dynamics computation

```

1. Input joint trajectory (therapy motion) for each joint  $i$ :  $q_i, \dot{q}_i, \ddot{q}_i$ 
2. initialize parameters: fixed-base (no 6D motion of the wrist) and gravity
   acceleration acting on the carpal (along x-axis, cf. Fig. 3).
3.  $V_0 \leftarrow \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}^T$ ,
4.  $\dot{V}_0 \leftarrow \begin{bmatrix} 0 & 0 & 0 & -9.81 & 0 & 0 \end{bmatrix}^T$ 
5. Forward recurrence: spatial velocities and accelerations propagation from
   carpals to distal phalanges:
for  $i = 1 \rightarrow n$  do
  (1)  $V_i \leftarrow R_{i+1,i}^T P_{i,i+1} V_{i-1} + H \dot{q}_i$ 
  (2)  $\dot{V}_i \leftarrow R_{i+1,i}^T P_{i,i+1} \dot{V}_{i-1} + \dot{P}_{i,i+1} R_{i+1,i}^T V_{i-1} + H \ddot{q}_i + \dot{H} \dot{q}_i$ 
end for
6. if (payload at the distal phalange is included) do  $F_{i+1} \leftarrow$  payload force
else do  $F_{i+1} \leftarrow \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}^T$ 
7. Backward recurrence: spatial forces calculated from distal phalanges to the
   carpals.
for  $i = n \rightarrow 1$  do
  (3)  $F_i \leftarrow P_{i+1,i}^T R_{i+1,i} F_{i+1} + I_i \dot{V}_i + \dot{I}_i V_i$  //forces
  (4)  $\tau_i \leftarrow H^T F_i$  // joint torques
end for
Return  $\tau$ 

```

The potential behind the dynamics modeling approach depends on having EMG-based kinematics data that reflects smooth and precise finger motions. By considering nonlinear motion effects, our hand model can be used in future robotic-assisted hand motion therapies. Our upcoming work is oriented towards the use of the EoM described by Algorithm 1 as the mathematical model of the patient's hand. This approach will enable the proper definition and tuning of a nonlinear controller for driving the exoskeleton movements. Hence, the complete dynamics modeling approach is defined as follows:

$$\tau = M(q)\ddot{q} + C(q, \dot{q})\dot{q} \quad (4)$$

Algorithm 2. Forward Dynamics Computation

```

1. Input joint torques  $\tau$  calculated by Algorithm 1.
2. Input an initial condition for  $q, \dot{q}$ 
3. Compute the composite-rigid-body articulated inertial operator (mass
   Matrix) from inverse dynamics in Algorithm 1:
 $M(q) \leftarrow \tau = M(q)\ddot{q} + C(q, \dot{q})\dot{q}$  //bounded to  $\ddot{q} \rightarrow 1, \dot{q} \rightarrow 0$ 
4. Compute the force velocity-dependent terms from inverse dynamics in
   Algorithm 1:
 $C(q, \dot{q}) \leftarrow \tau = M(q)\ddot{q} + C(q, \dot{q})\dot{q}$  //bounded to  $\ddot{q} \rightarrow 0, \dot{q} \rightarrow 1$ 
5. Compute the forward dynamics (eq. (4))
 $\ddot{q} \leftarrow M(q)^{-1} (\tau - C(q, \dot{q})\dot{q})$ 
6. Numerical integration for  $\ddot{q}$  to evolve the system over time.
Return new  $\ddot{q}, \dot{q}, q$ 

```

Table 5

ROC numerical data: accuracy of the top performance classifiers.

	Gestures					
	1	2	3	4	5	6
<i>Fine k-NN (KNN1)</i>						
Accuracy	0.96	0.96	0.96	0.99	0.99	0.99
<i>Weighted k-NN (KNN4)</i>						
Accuracy	0.95	0.96	0.96	0.98	0.97	0.98

Eq. (4) comprises the steps followed in Algorithm 1 for computing the EoM. The expression is well-known as the inverse dynamics Newton-Euler formulation written in the canonical form [68–71], being τ the applied torques, $M(q)$ is the rigid-body composite inertia matrix and $C(q, \dot{q})$ are the velocity dependent force terms (nonlinear terms of the model). Isolating the joint accelerations yields: $\ddot{q} = M(q)^{-1} [\tau - C(q, \dot{q})\dot{q}]$. This expression is known as the forward dynamics solution [72–74]. By following the Walker and Orin method [63], the forward dynamics requires the solution of the inverse dynamics to obtain $M(q)$ and $C(q, \dot{q})$ under the proper boundary conditions applied to the equation. Algorithm 2 details this approach.

3. Results

By following the acquisition protocol introduced in Fig. 2, this section summarizes the experimental results obtained with machine learning methods for classification of the six hand gestures of interest in this paper. Furthermore, a comprehensive analysis is carried out to demonstrate how the gesture-based finger motion kinematics can be applied to the hand dynamics model to validate our approach to robot-assisted hand-motion therapies.

3.1. EMG-based classification of gestures

From the EMG signal events, 24 features in time and frequency domains were computed, and used to test a total of 9 classifier configurations: 4 based on Support Vector Machines (SVM) and 5 based on Neural Networks (NN). First, the classifiers were trained with 1680 events, in order to find the optimal parameters that minimize the error function between the estimation and the real label of each event. Then, the performance of the methods was evaluated for 720 events in the testing set (cf. Fig. 2).

In order to compare the classifiers, a confusion matrix (6×6) was constructed, containing the number of events correctly and incorrectly classified per gesture. The sensitivity, specificity and accuracy for the methods were calculated, as the true positive, true negative and trueness rates, respectively.

Furthermore, Receiving-Operating Characteristic (ROC) curves were constructed by plotting sensitivity (S) versus 1-specificity (FPR) for the 9 aforementioned classifiers, as can be seen in Fig. 4. The average classifier performances for all gestures were between 0.69–0.93 of sensitivity and 0.01–0.06 of 1-specificity. In particular, the classifiers KNN1 and KNN4 had the best performance, with accuracy values as shown in Table 5. Moreover, the best classification on average within all classifiers was obtained for gesture 6 (sensitivity 0.97; 1-specificity 0.01) and the worst for gesture 1 (sensitivity 0.92; 1-specificity 0.06). Overall, we obtained an average sensitivity of 92%, specificity of 99% and accuracy of 98%, for the best classifiers.

3.2. Motion pattern analysis

In order to analyze the dynamics motion patterns of each EMG-based gesture, a 3D visualization of the model of the hand was developed. We used the Spatial Vector and Rigid-Body Dynamics Software Library developed by Featherstone [64,65]. Fig. 5 shows

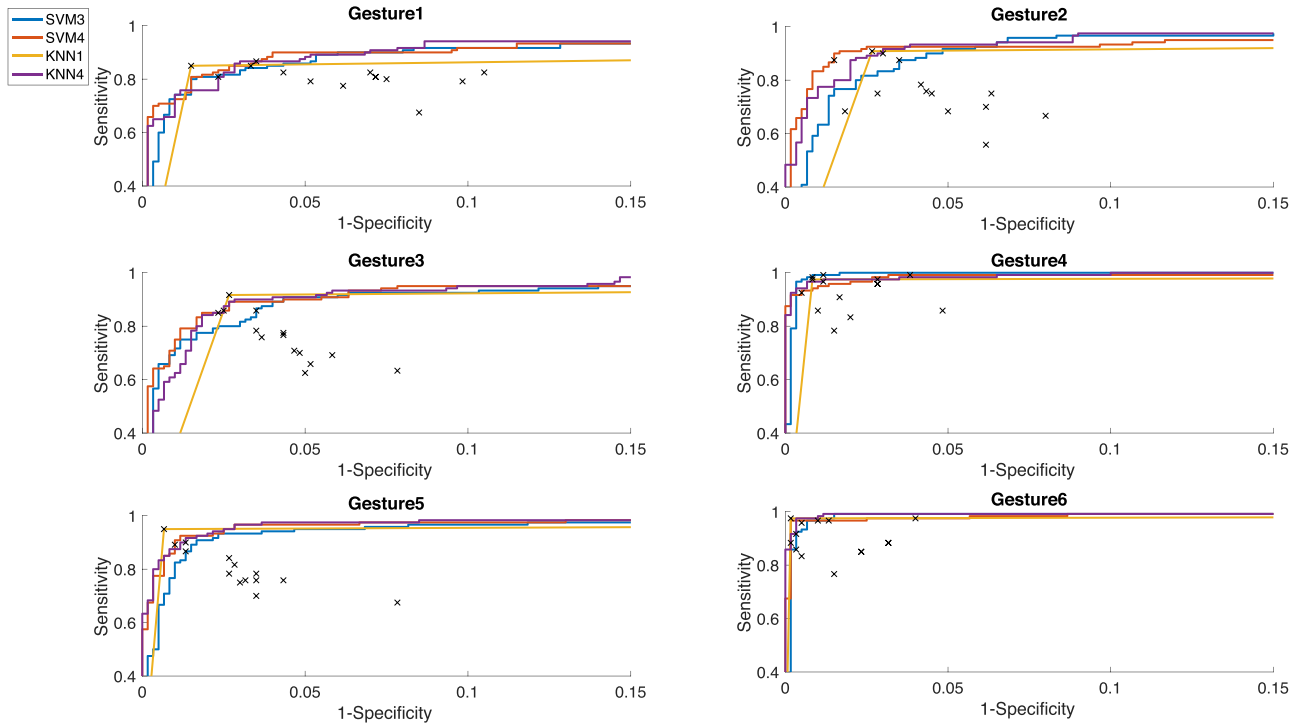


Fig. 4. ROC curves obtained for classification. In red, the performance of the best classifier on average for all gestures. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

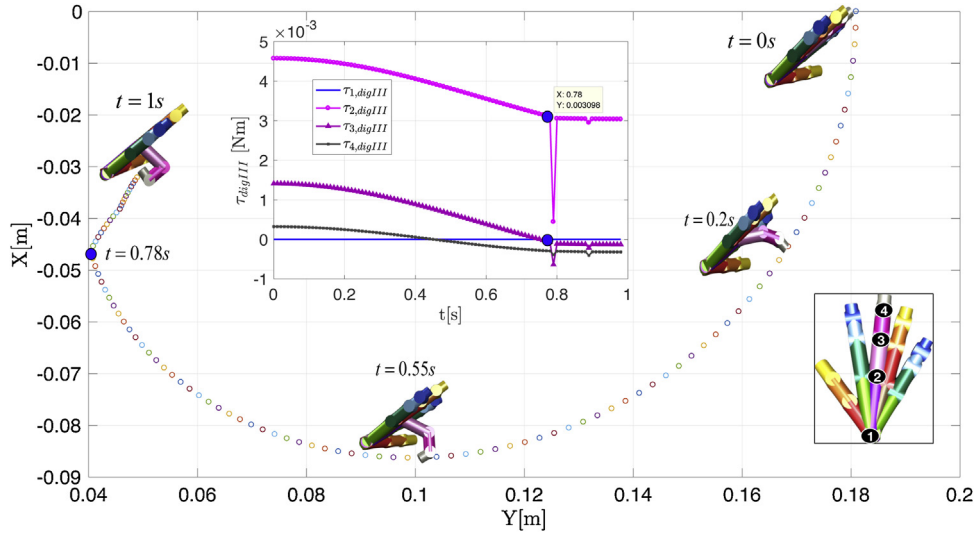


Fig. 5. Computational simulation for the middle finger: cartesian trajectory of digit 3 (gesture 2 from Fig. 2). Joint profiles ($q, \dot{q}, \ddot{q}$) are parametrized using EMG-based kinematics information of gesture 2. Using this joint profile as input to Algorithm 1, the corresponding joint torques are calculated, as displayed in the upper inset. The lower inset shows the four joints along digit III. Therefore, by using the joint torques as inputs to Algorithm 2, the cartesian trajectory for the tip of digit 3 is dynamically reconstructed.

the cartesian trajectory followed by the middle finger (digit 3) corresponding to the gesture 2 acquired from one individual (cf. Fig. 2). Using the joint profile as input to Algorithm 1, the inverse dynamics computation allows for quantifying the joint torques required to reproduce the motion according to the kinematics of gesture 2, as shown by Fig. 5. Subsequently, the output torques are then used as inputs to Algorithm 2 with the aim of reconstructing the physical cartesian motion described by the distal phalanx of the digit. Simulation tests applied to the other digits and gestures follow similar results.

Simulation results from Fig. 5 indicated that in general, smooth and precise motion for each digit can be achieved with interpolation

and parametrization of EMG-based gestures. However, depending on the gesture, small changes in the joint velocities might cause abrupt changes in the torque profile. In particular, our dynamics models are sensitive to these small variations of nonlinear motion effects, such as gyroscopic forces and Coriolis accelerations. This allows for the identification of abrupt changes during the digit motion, as observed in the torque peaks presented at $t = 0.78$ s. These non-desired effects can be counteracted by developing closed-loop controllers tuned by using the dynamics models presented herein.

Furthermore, a ground-truth was setup to measure the kinematics of each finger while the individuals were performing a

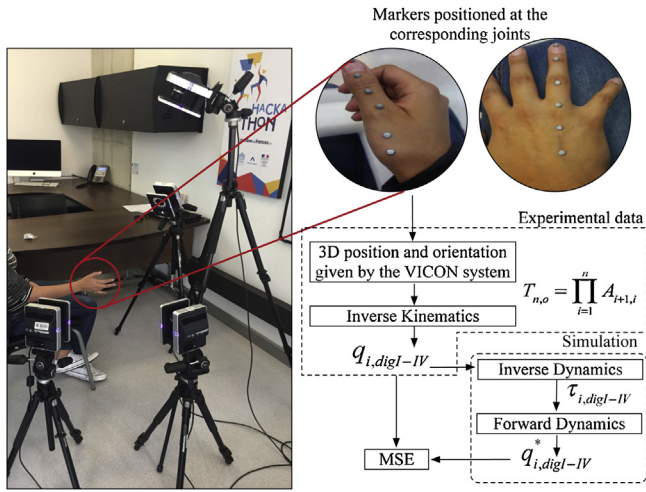


Fig. 6. Test-bed for hand motion pattern validation using a VICON system. Experimental results for the thumb tracking is shown in Fig. 7.

corresponding gesture. The forthcoming experiments have been conducted to validate the approaches for EMG-based joint trajectory generation and the dynamics modeling of the hand. The testbed setup was conformed by a 4-array VICON camera system, as shown in Fig. 6. Six markers were placed along the finger's joints in order to extract and measure the spatial kinematics of each gesture. Since the VICON system provides the three-dimensional motion of

each marker with a position resolution of 1 mm, we applied the DH-based forward kinematics approach to collect and organize the VICON data in the form of a 4×4 homogeneous-transformation matrix $A_{i+1,i}$. From Eq. (2), we calculated the orientation and position of consecutive joint frames (relating joint i with joint $i+1$). By applying the formula: $T = \prod_{i=0}^{n-1} A_{i+1,i}$, the cartesian trajectory of the finger's tip was represented by the operator T . Finally, the trajectory for each joint was calculated by applying a Jacobian-based numerical inverse kinematics method of the form: $q_k = J(q)^{-1} [T_k - T_{k-1}] + q_{k-1}$. The subscript k indicates the sample time of the variable.

Fig. 7 shows experimental results to validate our dynamics framework by analyzing the thumb's motion dynamics. Plot (a) depicts the experimental data captured by the VICON system. The markers were located at 4 joints along the thumb's finger as depicted in the inset. Trajectories for each joint were extracted and properly framed by following the DH-based kinematics method. Plot (b) highlights the thumb tip motion, denoted as joint q_4 from plot (a). The Cartesian path followed by the thumb's tip during flexion captured by the VICON (right) is compared against the simulation result obtained from the computation of Algorithm 2. After several trials, the average mean squared error between the VICON-based kinematics and the motion obtained by our dynamics models was $MSE = 0.0033$.

Finally, Fig. 8 shows simulation results that illustrate the steps performed by Algorithms 1 and 2. Plot (a) shows the corresponding joint trajectories used as inputs to our dynamics models. Plot (b) compares the joint accelerations obtained from the experimental kinematics data (upper plot) against the estimation of joint accel-

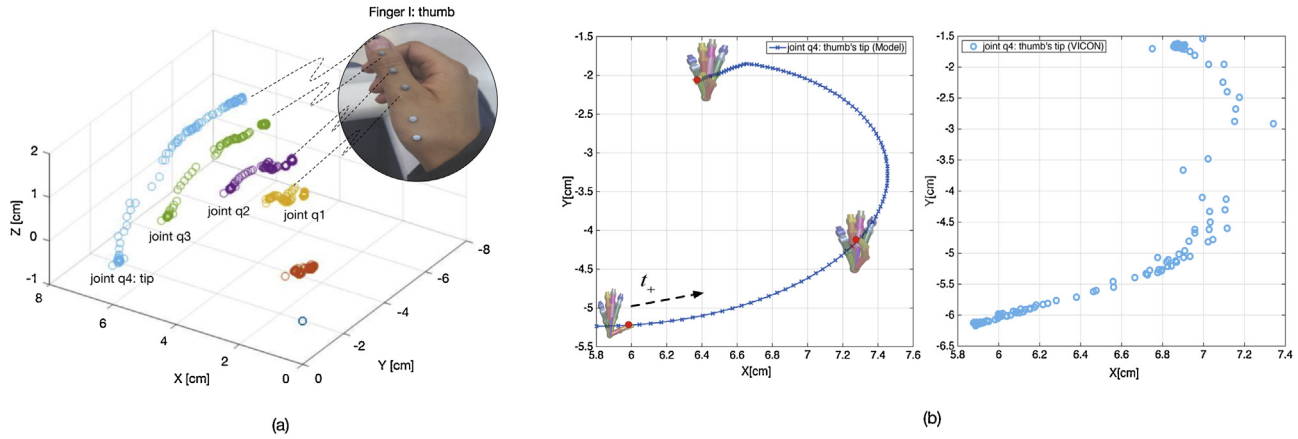


Fig. 7. (a) VICON experimental spatial mapping of the thumb's joints (4 DoF). (b) Dynamics model validation: the left plot shows the Cartesian motion for the thumb's tip flexion-gesture computed from Algorithm 2. The right plot shows the Cartesian motion for the thumb's tip from the experimental data captured from the VICON system.

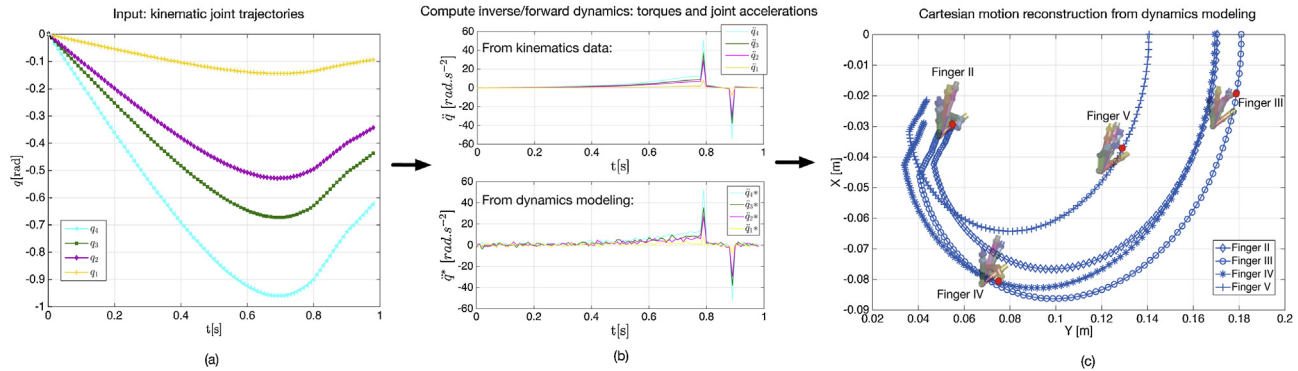


Fig. 8. (a) Input joint trajectories, (b) The top plot displays joint accelerations of reference (used as inputs to Algorithm 1 for computing the joint torques). The plot below displays joint accelerations computed from Algorithm 2; motion dynamics and white noise included during the estimation. (c) Dynamic computation of Cartesian trajectories followed by the tip of fingers II, III, IV, and V (thumb not included) while performing the corresponding gestures.

ations computed from the forward dynamics in Algorithm 2 (lower plot). On average, we have obtained a correlation coefficient of 0.91 between both values of $\dot{q}$. Plot (c) shows the motion simulated for each finger in the hand while performing a flexion gesture.

4. Discussion

Our experimental protocol, on 20 control voluntaries, consisted in 20 repetitions of 6 gestures. In total, we had 2400 events, or 400 events per gesture. The sample size was sufficient to demonstrate a good performance of the methods. However, for a clinical study, a larger database would be necessary. In particular, EMG signals from stroke patients should be analyzed in detail, as they could have an altered shape and a more variable behavior [38].

The EMG signals were acquired from the subject forearm with 4 electrodes. Whereas the recommended position of electrodes for finger movement recognition was reported in other works, we faced some difficulties to place them identically for all voluntaries. For instance, we found that a slight shifting of the electrode position allowed us to record a higher amplitude signal more related to the movements. However, in contrast to other studies with similar classification results, we were able to decode finger motion from with a lower number of electrodes.

A total of 24 features per event were extracted from EMG signals, which have been widely used in literature for movement classification. We did not implement a dimensionality reduction of the feature space, such as PCA, because we only had 24 features. Furthermore, the computation time for the feature extraction stage was less than 0.5 s. In a future study, it could be interesting to test for redundancy in the feature space and how this affects classification.

We have used gesture recognition to define the motion of a robotic-hand virtual model. This dynamic model plays an important role for analyzing how the extracted EMG-based motion patterns could be used in future robot-assisted therapies in stroke rehabilitation. An EMG-driven robotic system, e.g., an exoskeleton, would require the development of robust control methods to properly achieve precise, accurate and smooth motion assistance during the therapies. In this regard, our dynamics framework has been derived to be sensitive upon small variations of joint velocities, accelerations and moments (cf. Fig. 5).

5. Conclusion

In this work, we have used machine learning algorithms for classification of hand gestures, which defined the joint trajectories of a hand-model. This approach is interesting, as it could be applied to the field of robotic-assisted rehabilitation, in order to control a hand exoskeleton with EMG signals. In fact, EMG signals result of particular interest as they represent the subject's intention of movement. However, this field still presents different challenges and limitations. In particular, many proposed systems focus on rehabilitation of broad movements of the hand, instead of dexterous movements of the fingers. In addition, many of the proposed methods that use EMG signals for control of the exoskeleton, use a high number of electrodes or multi-channel systems. Furthermore, the validation of the outcome for robotic-assisted rehabilitation, compared to conventional therapy, still needs more evidence and is not conclusive. Further systematic clinical studies have to be done to demonstrate its effectiveness.

Here we have presented a first approach that aims to overcome some of the challenges. First, we decoded 6 hand movements, including individual finger movements. Second, we used only 4 electrodes in flexor and extensor muscles of the forearm. Then, with this information, we derived the joint trajectories and equations of motion for a virtual model of a hand. In the future, we would apply these results to a robotic exoskeleton. Specifically, we would

use the information of the EMG signals to adapt the performance of the robotic system, as long as the patient improves during the rehabilitation process.

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Conflicts of interest

The authors declare no conflicts of interest.

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